Honda's Asimo can also be your tour guide

That was 2001. Since then, there hasn't been much progress—and developers in labs around the world are still struggling to build bots that are spitting images of humans in movement and possibly behaviour. But when will they be ready? Will we, in our lifetime, ever be able to summon someone like Rosie—the robot maid from *The Jetsons*?

Rough estimates based on developments indicate that 2010 could well be the year when robots walk the street. Considering that's just five years away, it does seem rather impossible. They were also saying back in the '60s that everyone would fly by 2000 and that cars would be obsolete!

But what is it that we are looking to create here? Is a lump of wires and metal that vaguely resembles the human form the final objective? Not quite. Researchers around the world have since long achieved the human form—bipedal (two-footed) movement included—and work is now focused more on making these bots behave more like humans and understand what we are saying. Artificial Intelligence (or in fact, the lack of it) has been the stumbling block.

Coming Of Age

There are several things that differentiate humans from relatively simple creatures such as lizards, cockroaches and so on. Consider our sense of sight and our emotions.

Cockroach-like robots are relatively easy to build. But it's a tremendous challenge to create a bot that even vaguely resembles a human in the departments that matter. For instance, how do we imbue a robot with the emotion of fear? And how do we enable it to distinguish between a living and a non-living thing—something even a child can do when it is a year old?

Humans have normally overcome most of the challenges they have faced—and successfully built skyscrapers and supercomputers, and have even put men on the moon. But getting a robot to smell something, feel and identify, touch, see and imbue motor skills similar to humans is a task that is comparable only to the creation of a whole new world.

But why are emotions for a robot so important? Simply because for robots to be meaningful companions, they must be 'socially savvy'. At the MIT's Media Labs, work is being done on 'Kismet' (see box below), who is learning to recognise human emotions—and also has a face to express its own moods.

The image of a genie that has long been imbibed into the human psyche makes us want something—or someone—who can do more than us in terms of memory and mental and physical skills. Is a robot companion the answer?

The industrial world wants robots for work in hazardous surroundings; the military, to fight on the battlefield. But one place most humans would want them is in their homes. Why? For doing mundane, daily tasks like cleaning the house, washing clothes and cleaning the dishes. Driving you to work and shopping for groceries would figure next on the list, and an added bonus would be if they could tutor the kids after school. That's the ideal home AIO (all-in-one).

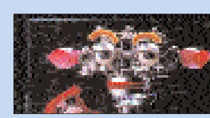
That's your genie right there, but given the state of AI and the current expertise in this field, getting your dream bot home may take some time. As of right now, they are prone to mistakes and a lot of them. But wouldn't that make them that much more human? Yes, but that's not what we're looking for, is it?

None of today's robots can do any of the household chores we would want them to. They

Growing Up Botty

MIT Labs had two humanoid robots to its credit. Unfortunately, both have retired and are now part of the MIT's Museum. During their 10-year lifecycle, these robots successfully mimicked emotions, and showed that it was possible to really 'grow up a bot'.

Kismet



Kismet was an expressive robot that could communicate with humans socially. It was capable of showing emotions such as sadness, happiness, excitement, and so on. A lot of computers worked together so Kismet could function socially at the level of a young person!

Kismet really could understand the human it was interacting with, and it processed the inputs it received (visual and auditory) from the human and adjusted its behaviour accordingly. It was designed such that it could learn the way an infant learns, and to this end, it needed a caretaker!

The caretaker had to provide a proper 'learning environment' for Kismet, and also understand how Kismet was 'feeling' at any given moment, so as to behave accordingly with the bot. Just like a child: if it was under-stimulated, Kismet acted bored—and was difficult to interact with. If over-stimulated, it showed fear, and if the excess stimulation continued, it shut itself down at some point.

Cog

Cog was an experiment in robotic cognition. The goal of the Cog project was to get it to the 'IQ' of a three-year-old. This humanoid torso had been learning to interact with its surroundings—including people—since its inception in 1993. Cog's intelligence levels progressed to the crawling-infant stage. Its 'eyes'—cameras, of course—tracked people moving, and it could establish eye contact with people. This may not seem like much, but consider the fact that most general robots can't distinguish a person from an object such as a wall.